

Incorporating HIIT Into a Concurrent Training Program

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To maximize performance across a number of sports, most athletes require not only high levels of cardiorespiratory fitness but also aspects of enhanced maximal strength, power/rate of force development, and speed (50), with varying degrees of these traits needed across different sports (see figure 1.6). For example, enhanced maximal strength and power/rate of force development is associated with improved performance in sport-specific skills, including sprinting, jumping, and change of direction (see reviews by 23, 95). Resistance training can also enhance both short- and long-duration performance (often termed *endurance*), due largely to neuromuscular adaptations that improve exercise efficiency (87), as outlined already in chapters 3, 4, and 5. It is therefore common practice for athletes to engage in some form of resistance (i.e., strength) training in combination with training aimed at enhancing cardiorespiratory and metabolic fitness (e.g., high-intensity interval training, HIIT). From this perspective, concurrent training represents an additional piece of the programming puzzle that practitioners must consider when prescribing endurance training modalities, including HIIT, with athletes.

The simultaneous integration of both strength and endurance training into a periodized training regime is termed *concurrent training*, a topic that has received considerable scientific interest since the first concurrent training studies appeared in the literature (47,

48). While concurrent training provides clear benefits for athletic performance by satisfying both ends of the strength-endurance continuum (77), this approach also presents additional logistical challenges to the practitioner. In particular, a primary concern with concurrent training is the potential for endurance training to interfere with adaptations primarily associated with strength training (i.e., enhanced maximal strength, power or rate of force development, and muscle hypertrophy) (for reviews see 6, 36)). A number of studies have indeed shown that, relative to performing strength training alone, concurrent training compromises gains in maximal strength (e.g., 9, 30, 34, 38, 41, 46, 47, 53, 57), aspects of power and rate of force development (38, 41, 46, 53, 57, 61), and muscle hypertrophy (9, 57, 82). While it is clear that endurance training may compromise aspects of strength training adaptation, the majority of evidence suggests that concurrent training does not attenuate improvements in $\dot{V}O_2\text{max}$ compared to endurance training performed alone (100). Taken together with the aforementioned evidence that strength training positively influences both cycling and running performance (87), these observations demonstrate the predominantly unidirectional nature of the interference effect. Put simply, endurance training may limit strength and power development during concurrent training, but strength training should not compromise endurance capacity

and instead will likely enhance it (figure 6.1). This does not mean that inclusion of strength training in any weekly training plan does not pose challenges for the programming of HIIT and other forms of metabolic conditioning; as will be discussed later in this chapter, there is also the potential for strength training to influence performance in the opposing mode (as well as vice versa).

It is important to note that the concurrent training field is rife with conflicting evidence regarding the

existence and nature of the interference effect to strength training adaptations. For example, not all studies suggest endurance training (referred to hereafter as *metabolic conditioning*) causes any interference to strength training adaptations during concurrent training (e.g., 7, 26, 70, 78). When an interference effect with concurrent training is observed, there is evidence for certain aspects of strength training adaptation to be compromised more than others. For example, attenuated training-induced improvements

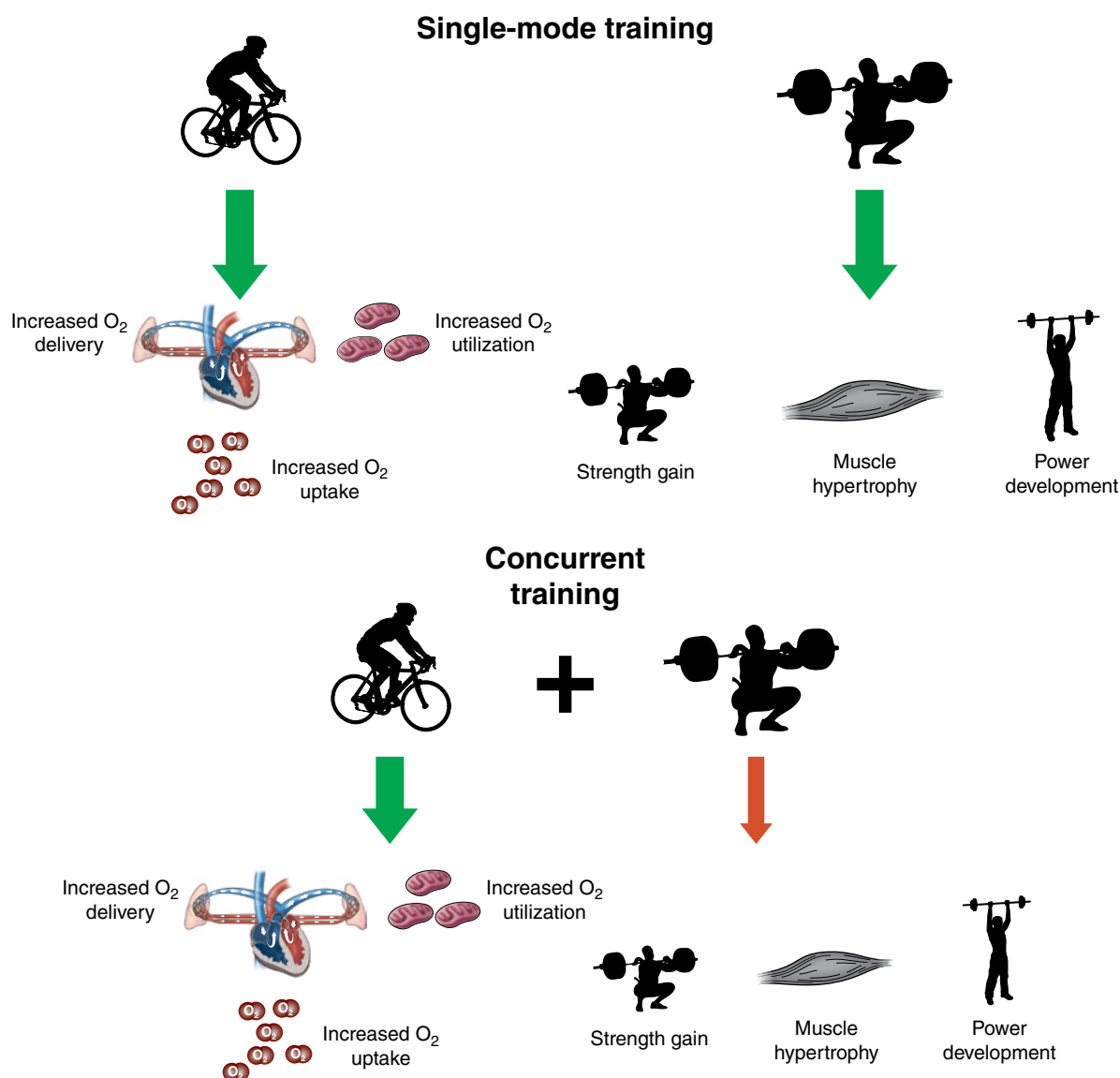


Figure 6.1 The interference effect illustrated. When performed separately, endurance and strength training lead to quite specific long-term adaptations. When performed concurrently, metabolic conditioning may limit strength and power development; however, strength training is unlikely to reduce (and instead will likely enhance) endurance performance (i.e., average power or speed maintained over prolonged durations or the ability to repeat high-intensity actions over time).

in power or rate of force development may occur without any measurable interference to strength gain or muscle hypertrophy (41). Interestingly, there is evidence that short-term concurrent training may even potentiate improvements in muscle hypertrophy (but importantly, not strength gain) (56, 66) versus strength training performed alone. The conflicting evidence base on the existence and magnitude of any so-called interference effect or concurrent training effect illustrates the complexity of the concurrent training field. These observations also suggest the interference effect is likely influenced by the specific manipulation of both training and nontraining (e.g., training status, nutrient availability) variables within the concurrent training program (36). As concurrent training involves the prescription and integration of both metabolic conditioning and strength training into the overall program, key training variables naturally include those related to the metabolic conditioning component (e.g., intensity, volume, type, modality), the strength training component (e.g., intensity, volume, type), and additional between-mode considerations (e.g., order in which endurance and strength training are performed in succession, length of between-mode recovery, nutritional intake around training sessions). When prescribing concurrent training, the programming puzzle challenge for the practitioner is to manipulate these variables to minimize any interference effect to strength training adaptations, while simultaneously maximizing improvements in cardiorespiratory fitness via metabolic conditioning. This then begs the question: how exactly do we minimize the interference effect when prescribing concurrent training? To answer this important question, and for practitioners to employ an evidence-based approach to limiting the interference effect during concurrent training, an understanding of the key factors responsible for the interference effect is required.

Potential Mechanisms for the Interference Effect and Ways to Minimize Possible Interference During Concurrent Training

Despite published evidence for the interference effect with concurrent training first appearing in the early 1980s (47), the mechanisms responsible are still incompletely understood. Although the exact factors are unclear, the interference effect likely is caused by:

- a compromised quality of strength training sessions, and/or
- a blunting of post-exercise processes that underpin aspects of strength training adaptation (e.g., attenuated post-exercise increases in rates of muscle protein synthesis and/or exacerbated rates of protein breakdown).

A compromise of either of these factors will likely result in suboptimal strength training adaptation. Variations of these two theories have been termed the *acute* and *chronic* hypotheses for the interference effect, respectively (24, 63).

The acute (or neuromuscular) hypothesis for the interference effect is based on the premise that metabolic conditioning simply limits the ability of the athlete to successfully perform strength training at the desired intensity and/or volume, consequently limiting strength training adaptation. In other words, metabolic conditioning simply ruins the quality of strength training sessions, leading to suboptimal development of strength, power or rate of force development, and/or muscle hypertrophy. In practice, it is easy to imagine how heavy strength training (e.g., squats and deadlifts performed at $\geq 85\%$ 1RM) is considerably more challenging to perform both at the required intensity (i.e., load) and for the desired number of repetitions or sets at a given load when the lower-body musculature is prefatigued from metabolic conditioning performed in the hours or days prior. If this situation is repeated over time, training-induced gains in strength, power, and/or muscle mass may be less than if strength training were performed in isolation. It is important to consider, however, that while strength training does not appear to inhibit long-term improvements in endurance performance, there is the potential for the residual effects of strength sessions to negatively influence metabolic conditioning sessions performed subsequently, for example, by influencing running kinematics and reducing economy (32). The practitioner must therefore be mindful of the potential for both strength training and metabolic conditioning to influence session performance in the opposing exercise mode, and that in some circumstances, performance during key HIIT or strength sessions may need to be prioritized over the other.

An alternate (but not necessarily mutually exclusive) theory for the interference effect has been termed the chronic (or molecular) hypothesis. This theory is based largely on the specificity of training adaptation and the apparent incompatibility between

large muscle fiber sizes and high levels of oxidative (endurance) capacity (98). Clearly, strength training and metabolic conditioning lead to quite specific long-term adaptations, although initially these adaptations may be less mode-specific compared to that seen following long-term training (22). In previously untrained individuals, for example, strength training can also improve $\dot{V}O_2$ peak (48), while aerobic exercise (specifically cycling, but not running) can promote type I muscle fiber hypertrophy (43, 44). This concept also brings us back to the toothpaste theory (initially described in chapter 4) when we discussed aspects of exercise prescription for elite athletes. While most forms of exercise do a decent job of squeezing from the middle of the tube (i.e., gaining some level of physiological adaptation and training outcome), after some time, it becomes an ineffective strategy to get the toothpaste out. Squeezing strongly and consistently on just one side before the other (i.e., a proper strength workout for strength development before appropriate HIIT for metabolic conditioning) and not squeezing just from the middle allows a more efficient means of dispensing toothpaste on our brush. Indeed, the higher the training status of an individual (and the less toothpaste in the tube), the more important it is that we squeeze our tube properly. At the extremes of the strength-endurance continuum, the specificity of training adaptation may be best illustrated by comparing the vastly divergent body types (i.e., phenotypes) of marathon runners to powerlifters, but similar differences can even be seen clearly within the same sport (i.e., sprint versus hill climbing cyclists and heavy versus lightweight rowers; chapters 15 and 16, respectively).

At the cellular level, the molecular factors responsible for this specificity of training adaptation in skeletal muscle are becoming increasingly clear (although these factors are certainly not yet completely understood). Adaptation to strength training (specifically, muscle fiber hypertrophy) has been primarily associated with the molecular factors governing muscle protein synthesis (i.e., the mechanistic target of rapamycin complex 1 (mTORC1) signaling pathway) (68), whereas endurance training adaptations in skeletal muscle have been linked to signaling pathways that regulate mitochondrial biogenesis (via activation of the transcriptional coactivator PGC-1 α , including the AMPK, CaMK, SIRT1, and p38 pathways) (51). It has been suggested that strength and endurance training stimulate a distinct

activation of these molecular signaling pathways within skeletal muscle, and that when repeated over time, these pathways largely underpin the divergent phenotypes induced by chronic training in either mode (consider marathon runners versus powerlifters as an example) (5).

Research investigating the molecular basis for the specificity of training adaptation has also provided clues to a potential mechanism for the interference effect with concurrent training. A classic study by Atherton and colleagues (5) aimed to replicate the stimuli associated with strength training and metabolic conditioning by electrically stimulating isolated rat muscles in either a strength training-like (repeated, high-intensity stimulation) or metabolic conditioning-like (prolonged, low-intensity stimulation) manner to investigate the molecular underpinnings of mode-specific training adaptations in muscle. The results suggested strength training-like electrical stimulation selectively activated the mTORC1 signaling pathway and increased the synthesis rates of contractile (i.e., myofibrillar) muscle proteins, whereas the metabolic conditioning-like electrical stimulation tended to activate only the AMPK/PGC-1 α signaling pathway, without any increase in the synthesis rates of contractile muscle proteins. These researchers also noted that the metabolic conditioning-like stimulation seemed to block the activation of the mTORC1 pathway, which may explain why myofibrillar protein synthesis was not increased by this type of stimulation. Based on these observations, it was suggested this so-called master-switch mechanism could potentially explain the specific adaptations caused in muscle by strength training and metabolic conditioning (5). This blunting of anabolic responses may also be explained by the ability of AMPK, which functions as a cellular energy sensor, to switch off the mTORC1 pathway in conditions of cellular energy stress such as those muscle cells experience during metabolic conditioning (54). In other words, when AMPK is activated by energy stress caused by endurance training (or at least endurance-like electrical stimulation), AMPK might act as a molecular brake on the mTORC1 pathway, which may blunt muscle protein synthesis rates and, in turn, potentially long-term muscle hypertrophy.

The next question, then, is can this master switch hypothesis explain why strength training adaptations (i.e., improvements in maximal strength, power, and muscle hypertrophy) may be blunted with concurrent training? Since the Atherton and colleagues (5)

study was published, many studies have investigated whether this molecular interference phenomenon actually occurs in human skeletal muscle following concurrent training. Despite the observations of the AMPK/mTORC1 master-switch mechanism occurring in rat muscle, at present there is virtually no evidence of a similar mechanistic function in human muscle during exercise. For example, in moderately trained individuals ($\dot{V}O_{2\max} = 361$ W), post-exercise anabolic responses in skeletal muscle after strength training were not impeded by prior HIIT cycling (5×4 min at $85\% \dot{V}O_{2\text{peak}}/3$ min at 100 W), despite prior HIIT increasing skeletal muscle AMPK activity ($\sim 90\%$) (3). There is also even evidence that anabolic responses (i.e., mTORC1 signaling) in skeletal muscle are increased to an even greater extent with concurrent training versus strength training alone (37, 65). Put simply, the ability of strength training to activate anabolic processes that regulate skeletal muscle growth appears not to be impeded in humans when preceded by a bout of strenuous metabolic conditioning, such as HIIT. It is also important to consider that metabolic conditioning itself is sufficient to activate anabolic processes (both mTORC1 signaling and increased rates of muscle protein synthesis) in human muscle (69), an effect that appears greater with higher-intensity metabolic conditioning (29). Additionally, training status is a key influencer of early post-exercise molecular responses in skeletal muscle (99), so further work is needed to investigate whether there is a molecular basis for the interference effect in more highly trained individuals.

When discussing the potential mechanisms underpinning the interference effect with concurrent training, it is perhaps important to consider that any molecular basis to the interference effect can likely only explain interference to muscle hypertrophy, as opposed to strength and/or power development. While muscle hypertrophy has traditionally been considered an important contributor to strength development (74), this notion is now being challenged (15), while it is also clear that increases in strength can occur with strength training in the absence of measurable changes in muscle size (101). The implications of any potential interference to muscle hypertrophy versus interference to strength development with concurrent training for the development of sport-specific physical capacities should also be considered by the practitioner. For example, while there is evidence that higher levels of strength

and power are associated with improved performance in sport-specific skills such as jumping, sprinting, and changes of direction (23, 95), evidence for the importance of muscle hypertrophy per se in developing these qualities is lacking. Because of the clearer links between strength and power development and sport-specific skill development, the implications of interference to strength and power development (as opposed to hypertrophy) with concurrent training are perhaps a more relevant consideration for the practitioner.

Nevertheless, while there is limited evidence of a molecular basis for the interference effect, it is acknowledged that training adaptation in skeletal muscle is largely mediated by the cumulative effects of early post-exercise responses to repeated training bouts (81). These include the molecular signaling events that underpin key adaptations to strength and metabolic conditioning, such as muscle hypertrophy and mitochondrial biogenesis (35). An understanding of the influence of concurrent training organization on early post-exercise responses, summarized for the reader in figure 6.2, may be potentially important for understanding the implications for long-term adaptations to concurrent training.

Linking the Metabolic and Neuromuscular Demands of HIIT With Their Potential Influence on Concurrent Strength Training

As discussed in chapter 4, prescribing HIIT requires the manipulation of at least 12 different variables, any of which may affect the short-term physiological responses to an HIIT session. Also described in chapter 5 was the concept that any given HIIT session will challenge, at differing respective levels, both the metabolic and neuromuscular physiological systems (chapter 3). In the context of prescribing HIIT in concurrent training programs, particularly important considerations include the neuromuscular demands of individual HIIT sessions (type 6 targets; figure 1.5) and how these demands may then influence performance during subsequent strength training sessions.

As discussed earlier in the chapter, an ideal scenario is whereby strength training sessions are undertaken when the neuromuscular system is less fatigued, so that strength training performance may

KEY POINTS

To summarize, prescribing concurrent training (as shown in most sports you'll find in the application chapters) presents unique challenges for the practitioner, given the potential for endurance training to interfere with the development of strength, power, and/or muscle hypertrophy induced by strength training. Although concurrent training likely benefits endurance performance over the long term, the potential for strength training to also compromise performance during key HIIT sessions should also be considered. Possible mechanisms for attenuated strength training adaptations with concurrent training include compromised strength training quality (i.e., the acute or neuromuscular hypothesis) and/or a compromised activation of anabolic processes after strength training sessions that underpin muscle hypertrophy (i.e., the chronic or molecular hypothesis; figure 6.2). While there is limited evidence for any molecular interference effect occurring in human muscle with concurrent training, there is certainly evidence that prior metabolic conditioning training can compromise subsequent strength training performance, as will be shown in the following section, lending support to the neural hypothesis for the interference effect. Because of the lack of evidence for any molecular basis for the interference effect, it is difficult to provide practical recommendations for minimizing

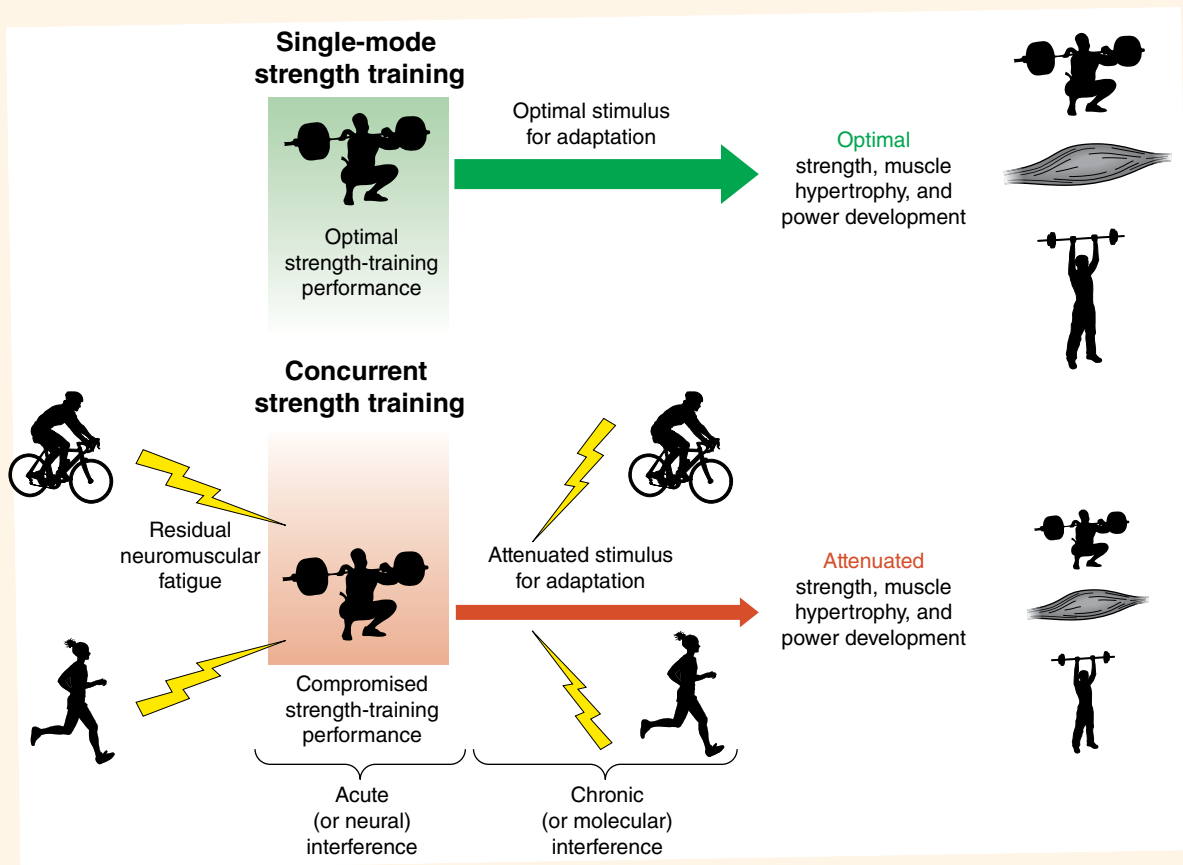
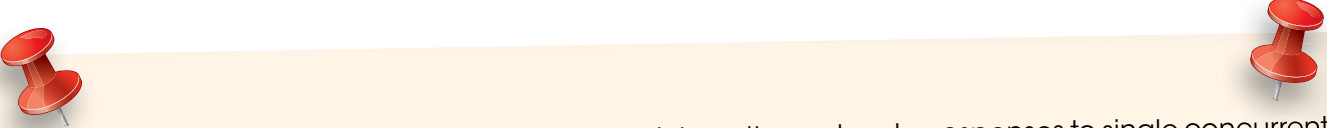


Figure 6.2 Conceptual framework outlining the potential mechanisms for the interference effect, including compromised resistance training quality and/or recovery from resistance training sessions. When strength training is performed alone (i.e., single-mode strength training), performance is maximized as minimal residual fatigue is present. This allows for an optimal stimulus for adaptation and, subsequently, optimal development of classic strength training adaptations (strength, muscle hypertrophy, power development). With concurrent training, the metabolic conditioning component likely causes residual neuromuscular fatigue that may compromise both strength training performance and potentially the adaptive response to strength training sessions. Both of these factors will likely limit the stimulus for adaptation, resulting in attenuated development of classic strength training adaptations.



interference with concurrent training based solely on the molecular responses to single concurrent exercise bouts. Rather, practical recommendations may be derived by considering factors related to metabolic conditioning sessions that are likely to compromise subsequent strength training performance, therefore likely leading to an interference effect. Practitioners integrating HIIT within concurrent training programs must therefore have a sound understanding of the physiological (anaerobic and neuromuscular in particular) demands of different HIIT formats (chapters 3 and 5) and how the residual effects of individual HIIT sessions may influence subsequent strength training performance, thereby influencing adaptation to concurrent training programs.

be maximized. This is because a less-fatigued athlete can likely:

- lift heavier absolute loads (at a higher percentage of his or her 1RM) for a given number of repetitions,
- perform more repetitions at a given percentage of the 1RM load, and/or
- perform repetitions at faster intended and/or actual concentric velocities (as a percentage of the maximum velocity possible with a given load).

These factors likely ensure the strength training stimulus (type 6, figure 1.5) is maximized, and therefore, the potential for adaptation to that strength training session likewise is optimized. As such, the challenge for the practitioner is to schedule concurrent HIIT and strength training sessions so that the residual effects of the HIIT sessions do not have a substantial negative influence on strength training performance.

The practitioner must consider that two HIIT sessions may similarly challenge the cardiopulmonary and oxidative systems, while at the same time involve distinctly different anaerobic energy contributions and/or neuromuscular strain (chapter 5). Any HIIT session that evokes a large neuromuscular strain component (i.e., types 2, 4, or 5) will likely result in more substantial residual fatigue of the exercised musculature, compared to HIIT sessions with smaller neuromuscular strain components (i.e., types 1 or 3). For this reason, HIIT sessions characterized by larger neuromuscular demands will likely further compromise subsequent strength training performance for a period of time after HIIT cessation, compared to HIIT sessions of lower neuromuscular demands. Similarly,

it is likely that strength sessions (type 6) may also compromise performance during subsequent HIIT sessions with high neuromuscular demands (i.e., types 2, 4, or 5). As you will see in later sections of this chapter, the practitioner must consider the relative priority of key HIIT or strength sessions within any given training week and manipulate the organization of these sessions to minimize the potential for negative interactions between training modes.

There is strong evidence that the force-generating capacity of exercised musculature is compromised in the subsequent hours following metabolic conditioning performed at various intensities (2, 10, 11, 60, 61). As discussed, residual fatigue from endurance training may compromise performance during a strength training session conducted within this period, primarily by limiting the number of repetitions possible at a given absolute or relative intensity, thereby limiting total session volume (28, 59, 83, 92, 96). This attenuation of within-session strength training performance is likely to, at least partially, mediate the interference effect during periods of concurrent training (24, 63).

From a practical perspective, it is useful then to consider the effects of HIIT variable manipulation on the neuromuscular demands of HIIT and, subsequently, the potential implications for subsequent strength training performance. Currently, however, there are considerable gaps in our knowledge surrounding the influence of HIIT variable manipulation on neuromuscular demands, and particularly the implications of these demands for performance during subsequent training sessions (e.g., strength training). Current understanding of the factors modulating the neuromuscular demands of HIIT are well described in chapters 3, 4, and 5; however, these factors are summarized for the reader in figure 6.3.

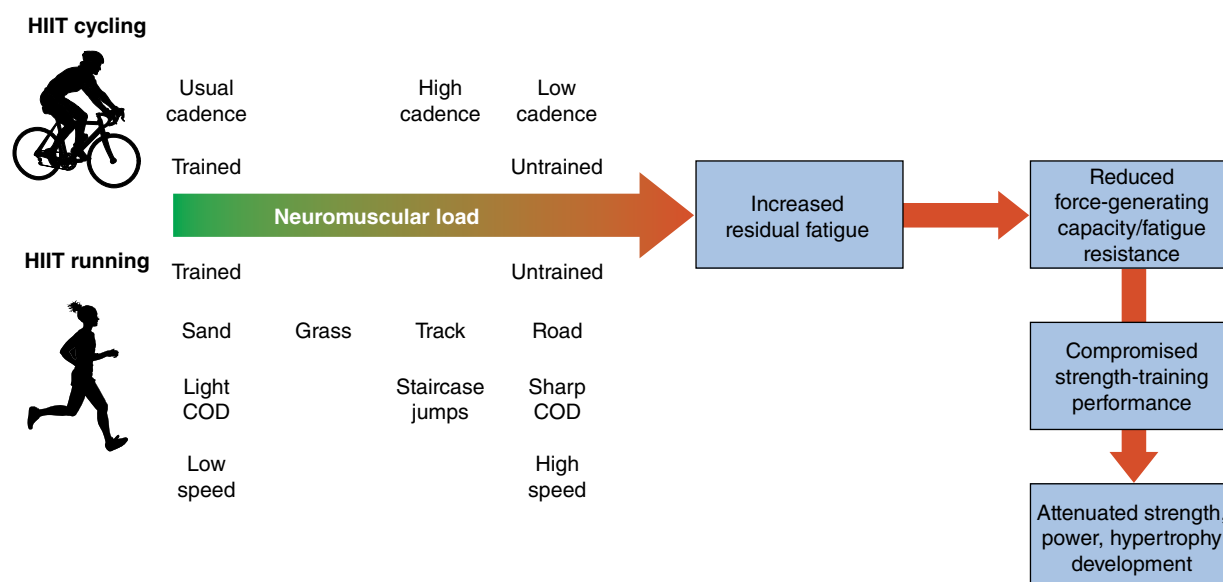


Figure 6.3 Conceptual framework summarizing the practical factors that might influence residual fatigue following HIIT, and therefore may further negatively influence subsequent strength training performance.

The factors outlined in figure 6.3 should be considered when integrating HIIT with strength training, given that residual effects of such metabolic conditioning examples may negatively influence strength training performance. Indeed, a number of studies (e.g., 28, 59, 83, 92, 96) have investigated the consequences of prior metabolic conditioning (of various relative intensities) on subsequent strength training performance. While most studies show a negative effect of prior metabolic conditioning on subsequent neuromuscular and/or strength training performance (28, 59, 83, 92, 96), others instead show no effect (62). These equivocal findings suggest that, like the interference effect itself, this phenomenon likely depends on the intricacies of the prescription of both training- and nontraining-related variables within a concurrent training program. The reality of these complexities is highlighted in the application chapters in part II, written by coaches and sport scientists charged with the daily delivery of evidence-based practice in elite sport. Although much further work is needed to identify optimal strategies for integrating HIIT into concurrent training programs, existing evidence has begun to shed light on the potential effects of training (e.g., HIIT work intensity, volume, and modality) and nontraining (e.g., within-session training order, length of between-mode recovery, and the potential

influence of training status or age) variables on subsequent strength training performance.

Effect of HIIT Work Intensity

Given the focus on HIIT, the potential effects of HIIT work intensity in modulating the interference effect with concurrent training is a highly relevant practical consideration. There is evidence that the compromising effect of metabolic conditioning on subsequent force-generating capacity and strength training performance may be somewhat intensity dependent (11, 28, 62). For example, neuromuscular fatigue (indexed by reduced isometric MVC force and quadriceps muscle activation assessed via EMG) of the exercised musculature is evident for at least 6 h after combined continuous/HIIT cycling (30 min at 80% $\dot{V}O_{2\max}$ followed by 4 × 60 s efforts at 120% $\dot{V}O_{2\max}$) in moderately trained triathletes ($\dot{V}O_{2\max} = \sim 59 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$) (11), whereas less residual fatigue is often noted following lower-intensity continuous endurance training (28, 62). It follows that the negative impact of metabolic conditioning-induced fatigue on strength training performance is worsened with higher-intensity metabolic conditioning (28, 59, 83). This finding is not universal, however, with detriments to strength training performance in recreationally active athletes

($\dot{V}O_{2\max} = \sim 57 \text{ mL}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$) shown to be similar when preceded by either submaximal continuous (36 min at 40% $\dot{V}O_{2\max}$) or HIIT (6 $\times$ 3 min at 95%-100% $\dot{V}O_{2\max}$ /3 min at 40% $\dot{V}O_{2\max}$) cycling (92). Additional work has also shown no difference between moderate-intensity continuous (30 min at 80% of the lactate threshold [LT]) and HIIT (10 sets of 1 min at 120% LT, 1 min passive recovery) cycling on either anabolic responses in muscle to subsequent strength training (37) or 1RM leg press strength gain after 8 wk of training (38). It is important to consider, however, that although these studies highlight the influence of relative endurance training intensity for eliciting residual neuromuscular fatigue and compromising strength training performance or adaptation, virtually nothing is known about the effect of different HIIT work interval intensities on these parameters. For example, whether HIIT work intervals performed at 120% versus 100% $\dot{V}O_{2\max}$ elicit greater neuromuscular fatigue, and how this influences the quality of subsequent strength training sessions, is currently unclear. Regardless of this lack of empirical evidence for the role of HIIT intensity in modulating neuromuscular fatigue, the more conservative option for the practitioner may nevertheless be to reduce HIIT intensity prior to strength training sessions and consequently favor HIIT types 1 and 3 as opposed to HIIT types 2, 4, and 5 (figure 1.5).

Effect of HIIT Volume

Although not specific to HIIT volume per se, there is some evidence that the total volume of endurance training is an important modulating factor for the interference effect during concurrent training (55, 88, 100). A meta-analysis of the concurrent training literature (100) suggested increased weekly frequencies and/or durations of endurance training sessions are associated with greater interference to strength training adaptations. Additional indirect evidence for a volume-dependent interference effect is provided by the absence of interference with lower endurance training frequencies (≤ 2 sessions/wk) (39, 41, 70), whereas attenuated strength development is observed with higher weekly endurance training frequencies (≥ 3 sessions/wk) (9, 46, 47, 55, 57). These observations suggest that strategies that limit the frequency and/or session volume of HIIT may be worthwhile for minimizing interference during periods of concurrent training, particularly during training peri-

ods during which the development of maximal strength, power, and/or hypertrophy may be prioritized. Despite this, much further work is required to better define the role of overall HIIT volume in mediating the interference effect.

Effect of HIIT Modality

As discussed in chapters 4 and 5, HIIT may be programmed using multiple metabolic conditioning modalities (e.g., running, cycling, swimming, rowing). It is important to recognize that different training modalities are characterized by unique patterns of muscle recruitment, as well as differences in the magnitude of neuromuscular strain imparted on the active musculature. For example, running and cycling primarily recruit the lower-body musculature, whereas both swimming and rowing involve a considerable element of upper-body muscular work. Despite these obvious between-mode differences in muscle group involvement, comparisons between the neuromuscular demands of different endurance training modalities, i.e., running, cycling, swimming, rowing (particularly in reference to HIIT) are currently lacking. Nevertheless, due to an increased eccentric loading component (71), running likely imparts greater neuromuscular strain on the lower-body musculature compared to cycling, whereas both swimming and rowing will clearly induce greater neuromuscular strain on the upper-body musculature versus either cycling or running. These factors have implications for the integration of HIIT within concurrent training programs because differences in muscle recruitment and neuromuscular strain profiles associated with different HIIT modalities, specific to the exercise mode, are likely to differentially influence performance in particular strength-training exercises performed subsequently. For example, running-based HIIT will likely further compromise subsequent lower-body strength training performance versus cycling-based HIIT, whereas upper-body strength training performance may be limited by both swimming- and rowing-based HIIT. It is also important to note that similar considerations can also be held for the potential influences of strength training on performance during subsequent key HIIT sessions, as discussed previously. While these observations likely provide some insight into the potential effects of different metabolic conditioning modalities on subsequent strength training performance,

there is very little published literature available to verify these insights. One study (79) compared the effects of HIIT cycling or treadmill running (15×1 min at $v/p\dot{V}O_{2\max}$ /1 min passive recovery) on subsequent strength training performance (4 sets of maximum repetitions of half squats at 80% 1RM) in recreationally active males ($\dot{V}O_{2\text{peak}} = \sim 45 \text{ mL}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$ and $\sim 38 \text{ mL}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$ measured during running and cycling, respectively) (79). Both HIIT cycling and treadmill running limited the number of repetitions performed and total volume load lifted (in kg) during subsequent sets of half squats, although cycling actually resulted in more prolonged negative effects (79). As the prior cycling- or running-based HIIT did not influence muscle activation (quantified via vastus lateralis EMG activity) during strength training, the authors speculated their findings were mediated by a higher reliance on anaerobic metabolism with cycling (i.e., HIIT types 3 and 5), although markers of this were not measured. It is possible these findings may have also been influenced by the training status of the participants with reference to running versus cycling. Based on the higher reported $\dot{V}O_{2\text{peak}}$ values during running (79), it is possible these participants were more accustomed to running versus cycling, which may have led to increased residual fatigue following cycling and, in turn, further compromised strength training performance. Clearly, more work is needed to confirm the effects of different HIIT modalities on residual neuromuscular fatigue and the associated implications for strength training performance.

Effect of Within-Session Order and Recovery Duration Between HIIT and Strength Training Sessions

As neuromuscular fatigue induced by metabolic conditioning is a transient phenomenon (10, 11, 92), the recovery duration allowed between endurance and strength training sessions likely influences the effects of prior metabolic conditioning on strength training performance (and vice versa). Most evidence suggests that neuromuscular fatigue persists for at least 6 h after strenuous endurance exercise (including HIIT), which may be fully resolved after 24 h (10, 11, 92). It should be kept in mind, however, that although often implemented by practitioners, the 6 h window may not always be the magic number for optimal between-mode recovery length, since the recovery rate is likely affected by a number of

modulating factors, including the type of exercise performed and an athlete's training status (i.e., figure 6.3). Nevertheless, extended recovery periods after HIIT sessions are likely optimal for resolving any residual fatigue (particularly following HIIT sessions involving large neuromuscular demands) and hence are likewise optimal for maximizing subsequent strength training performance. For this reason, it is also advisable that when strength training and metabolic conditioning are performed within a single training session, strength training should precede the metabolic conditioning component (rather than vice versa) (76). There is indeed some evidence that increasing the recovery periods between endurance and subsequent strength training can limit any negative effects on strength training performance. For example, completing either submaximal continuous (36 min at 40% $p\dot{V}O_{2\max}$) or HIIT (6×3 min at 95%-100% $p\dot{V}O_{2\max}$ /3 min at 40% $p\dot{V}O_{2\max}$) cycling compromises strength training (leg press) performance at both 4 and 8 h post-exercise, although these negative effects were resolved after 24 h (92). There is also some indirect evidence for the benefits of increasing between-mode recovery lengths when comparing strength gains to concurrent training when strength and metabolic conditioning sessions are performed on separate days, compared to being performed on the same day. For example, one study (89) showed that performing low-intensity strength training (15-20-RM loads) on different days than HIIT sessions ($6-8 \times 3$ min cycling bouts at 90%-100% $\dot{V}O_{2\max}$) led to superior 1RM leg press strength gains (but not markers of muscle hypertrophy), compared to undertaking both training modes on the same day. Additional evidence (20) in recreationally trained subjects ($\dot{V}O_{2\max} = \sim 40 \text{ mL}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$) suggests even supramaximal-intensity cycling (SIT involving $4-6 \times 20$ s all-out modified Wingate-style sprints) does not cause interference to maximal strength or hypertrophy responses when performed on alternating days to high-intensity ($\sim 85\%$ 1RM) strength training. Increasing the recovery duration between HIIT (3×6 min sets of 15 s 120% MAV/15 s passive recovery) and strength training (from 0 to 24 h recovery) seems to optimize strength and power development in amateur rugby players, even when HIIT sessions were always performed after strength training (86). Since HIIT was not performed prior to strength training in that study (86), it is therefore unlikely these results could be explained by compromised strength training performance. Interestingly

however, increasing the between-mode recovery duration also led to larger improvements in $\dot{V}O_{2\text{peak}}$ (86), which may reflect compromised HIIT performance when preceded closely by strength training (although this was not directly assessed). Indeed, the potential for residual fatigue from prior strength training sessions to negatively influence performance during subsequent HIIT sessions (as well as vice versa, as previously discussed), including altering factors such as running kinematics and economy (32), is worth considering. For example, single bouts of strength training can compromise subsequent cycling time trial (15 min) performance (17), and both running (31) and arm cranking (33) time to exhaustion. Performing strength training before endurance training can also compromise training-induced improvements in running time trial (4 km) performance, compared to the reverse exercise order (21). While these observations highlight the potential bidirectional interactions between strength and endurance training acutely influencing performance in the opposing mode, the weight of evidence nevertheless suggests strength training can, in the long term, improve various aspects of endurance performance without compromising $\dot{V}O_{2\text{max}}$ (1).

Aside from performance enhancement alone, considerations for the order of concurrent training sessions also extends to modulating injury risk during HIIT (64). This is because strength training (e.g., Nordic hamstring lowers) may transiently reduce levels of muscle strength and/or activation, which in turn may expose the musculature to increased loading during high-speed running sessions performed subsequently and potentially increase injury (e.g., hamstring strain) risk (64). In these cases, the practitioner should be mindful of programming high-speed running for a period after strength training sessions and manipulate HIIT formats accordingly to constrain absolute running speeds when performed subsequent to strength training.

Altogether, the available evidence suggests that where possible, strength training should be performed prior to HIIT if performed within the same session or at least 6 h after HIIT sessions if performed within the same day. The ability for strength training sessions to also induce neuromuscular fatigue and then influence performance (and also injury risk) during subsequent HIIT sessions performed must also be considered. Ultimately, the practitioner will need to decide on the relative priority of any given HIIT or strength training session and adjust the weekly training plan accordingly to ensure perfor-

mance is maximized within priority training session(s).

Role of Athlete Training Status or Age

As discussed in chapter 3, individual training status (that is, the athlete's fitness level or ability to perform a given exercise task) will largely influence the degree of neuromuscular fatigue induced by a given HIIT session, and therefore the likelihood of that session compromising subsequent strength training sessions. Interestingly, there is evidence that for some athletes, neuromuscular performance may be transiently enhanced, rather than compromised, following strenuous HIIT sessions (e.g., 14). This observation has been attributed to a possible postactivation potentiation effect (13), whereby prior high-intensity muscular contractions can acutely enhance performance during subsequent activities (49). The practitioner should therefore consider the possibility that prior HIIT sessions (likely those performed in close proximity to strength training and with a low-anaerobic contribution and/or associated muscle contractile perturbation, i.e., types 1 or 2) may actually enhance, rather than inhibit, subsequent strength training performance in highly trained athletes. It is also possible that in some circumstances, prior strength training sessions may also likewise enhance performance (e.g., power output) during subsequent HIIT sessions (18). However, the factors that modulate this potentiating effect are complex and likely related to the intricacies of both the type of HIIT session performed, the type of subsequent activity or strength training performed, as well as the training status of the athlete. Much further research is required to determine the role of these factors in modulating the influence of prior HIIT sessions on the capacity to acutely enhance or compromise performance during subsequent strength training sessions (as well as vice versa).

Overall it is clear that neuromuscular fatigue from metabolic conditioning can negatively influence performance during subsequent strength training sessions, and this may at least partially explain attenuated strength training adaptation with concurrent training. Understanding of the neuromuscular demands of different HIIT formats allows the practitioner to manipulate HIIT prescription in consideration of the potential consequences of these sessions for other training modalities performed within a weekly microcycle (e.g., strength training). While

current evidence has begun to shed light on the neuromuscular demands of different HIIT formats and the impact of metabolic conditioning (including HIIT) on subsequent strength training performance, this evidence is not without limitations, and our understanding of these factors is by no means complete. Although there is no doubt much further work is needed to identify optimal strategies to integrate HIIT into concurrent training programs, existing evidence has nevertheless begun to shape practical recommendations for minimizing the interference effect.

It should also be recognized that in some circumstances, the weekly periodization of strength training and HIIT may be dictated not by what is necessarily optimal but by other practical factors (e.g., coach preference with weekly schedule and/or within-session order of training content, priority of HIIT versus strength training quality, etc.). In some cases (particularly in endurance sports), the quality of key HIIT sessions may often need to be prioritized ahead of strength training sessions (18). In these situations, the practitioner may need to accept compromised strength training quality to allow for this emphasis on HIIT sessions.

Piecing It Together: Practical Strategies for Programming HIIT During Concurrent Training

As discussed throughout the chapter, programming HIIT during concurrent training requires consideration of how individual HIIT sessions may influence performance during subsequent strength training sessions and, in turn, modulate any potential interference effect to strength training adaptations. Although more research is required to evaluate optimal strategies for integrating HIIT within concurrent training programs, a number of practical recommendations for maximizing concurrent training adaptation may be devised based on our current understanding. It should be made clear, however, that these recommendations depend on multiple logistical factors (see part II), including the particular sport itself, the training phase, athlete training status, etc. Nevertheless, here are general recommendations for limiting interference to strength training adaptations with concurrent training.

- Limit neuromuscular fatigue prior to strength training sessions by performing strength training

either before or at least (approximately) 6 h after HIIT sessions.

- HIIT sessions performed in the 24 h prior to strength training sessions should ideally be characterized by lower neuromuscular demands by incorporating lower absolute or relative work intensities associated with lower neuromuscular load (i.e., types 1 and 3). However it is worth noting that using lower HIIT intensities may often need to be associated with longer effort durations (e.g., 3 min at 85% V_{IFT} versus 10 s/20 s at 120% V_{IFT}), and consequently these longer durations may also cause neuromuscular fatigue. There is, in fact, a likely optimum combination between exercise intensity and duration when it comes to neuromuscular responses and fatigue (figure 5.42).
- Limit prescription of all-out HIIT modalities (i.e., RSS, SIT formats, or in general, neuromuscular-demanding HIIT formats, i.e., types 2, 4, and 5).
- Performing running-based HIIT on softer surfaces (e.g., grass or sand instead of road) is likely to impart less neuromuscular strain (figure 5.43).
- Limit the change-of-direction (COD) component (particularly sharp COD) of running-based HIIT or employ cycling-based, rather than running-based, HIIT (depending on the athlete's training specificity).
- Where possible and when individual strength or power development (type 6) may be favored over the metabolic conditioning component, total weekly metabolic conditioning volume should be limited by reducing the weekly frequency of HIIT sessions and/or performing fewer repetitions and/or series during HIIT sessions or by reducing training volume associated with other cardiorespiratory modalities (e.g., prolonged endurance training sessions).

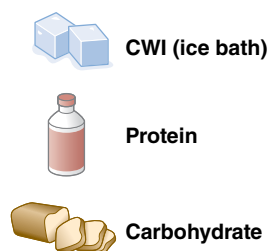
Based on these recommendations, a number of practical scenarios in which HIIT is integrated into concurrent training programs are shown in figures 6.4 and 6.5, with examples of optimal organization of HIIT and strength training within a weekly microcycle of team and individual sport programs, respectively. Excellent team and endurance sport examples of weekly microcycle periodization of concurrent training can additionally be viewed in nearly every chapter in part II.

Team sport weekly periodization example: preseason

Sub-optimal

1. High neuromuscular strain from type 4-5 HIIT will likely compromise subsequent strength session.
2. Type 4 SSG HIIT session will likely cause neuromuscular fatigue for at least 6 h afterward. Strength session likely compromised if performed within this time window.
3. Residual fatigue on day 4 might be worsened by type 2 session performed on previous day.

Morning	Off	SSG + skills	HIIT short (type 2)	HIIT short (SSG type 4 MW)	Off	SSG + skills Sub-max endurance	Off
	SSG + skills Sub-max endurance	HIIT RST/SIT (type 4-5) Strength	Off	Strength	Speed	Off	Off
Day	1	2	3	4	5	6	7



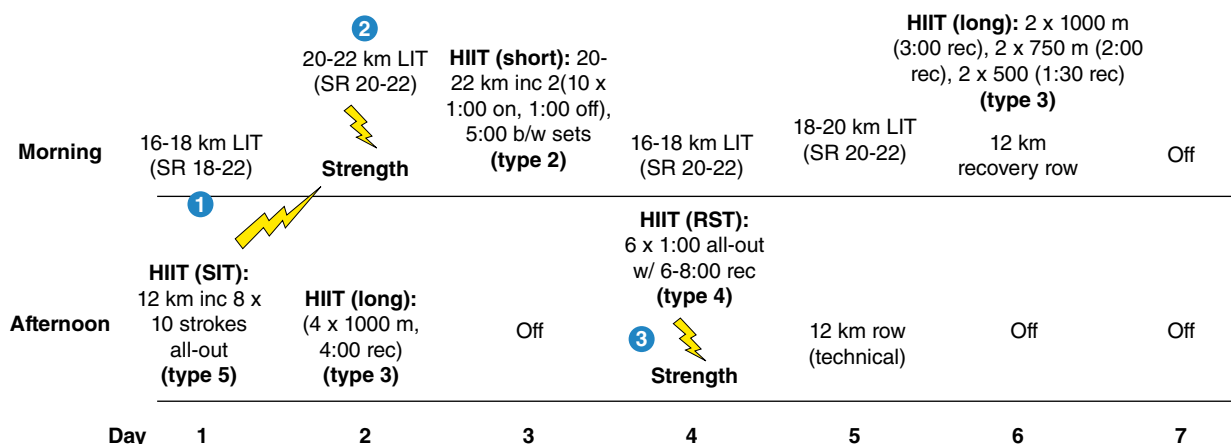
1. Strength session moved to AM on day 2 so it is performed with less neuromuscular fatigue. In this scenario, the practitioner should also be mindful that prior strength work may increase injury risk while running in later metabolic conditioning sessions. Running speeds should therefore be limited during these sessions to reduce injury risk.
2. Strength moved to AM on day 4, OR recovery duration extended to >6 h. If strength is performed in the PM after an AM HIIT session, ensure adequate recovery length and nutrient provision between modes.
3. Type 2 HIIT modified to type 1 HIIT to further reduce neuromuscular fatigue on day 4.
4. Sub-max endurance session moved to day 1 AM to avoid any potential negative influence on strength session performed the following morning.

Morning	SSG + skills Sub-max endurance	SSG + skills Strength	HIIT short (type 1)	HIIT (SSG type 4 MW) SSG + skills	Off	SSG + skills Sub-max endurance	Off
	Off	HIIT RST/SIT (type 4-5)	Off	Strength	Speed	Off	Off
Day	1	2	3	4	5	6	7

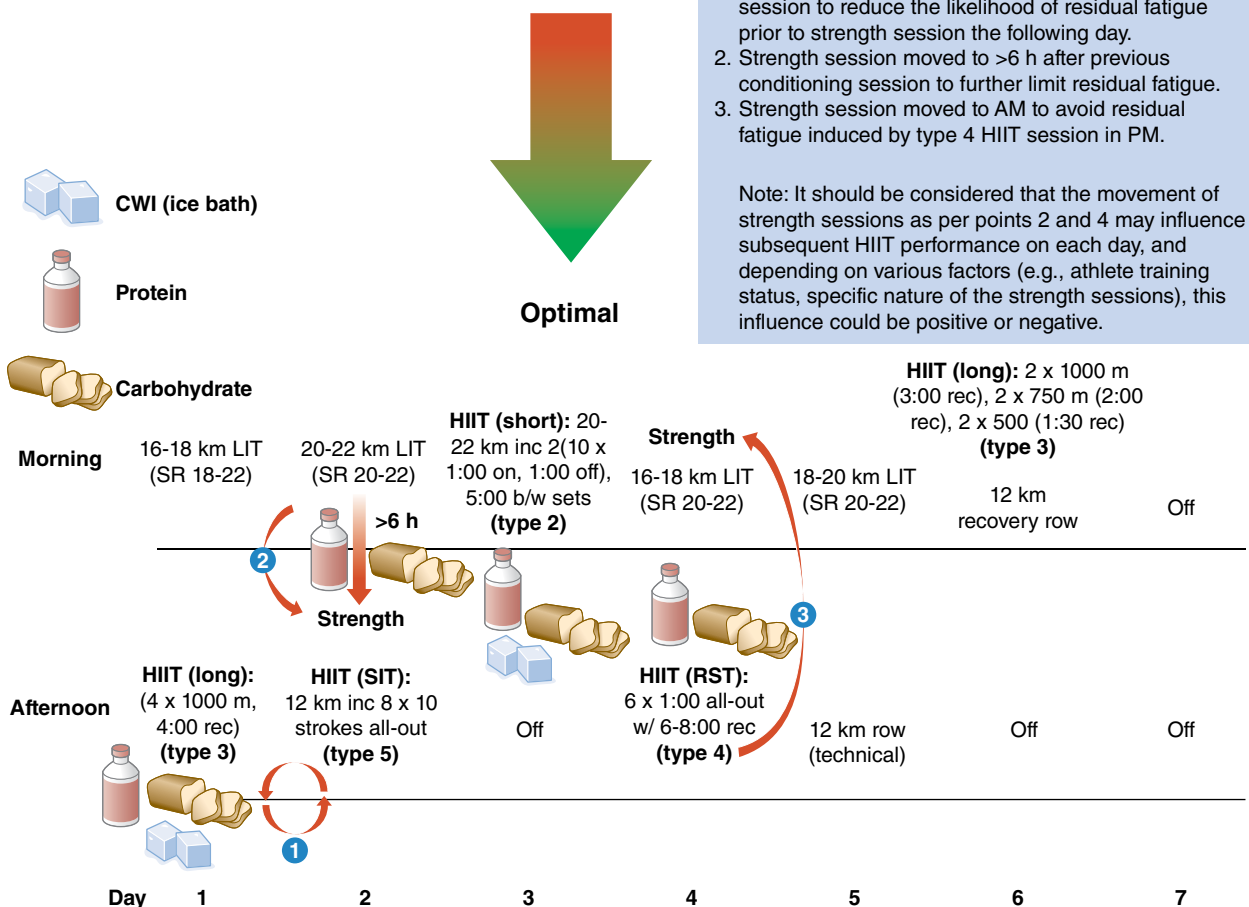
Figure 6.4 Team sport concurrent training program puzzle example.

Individual sport weekly periodization example: hard training week

Sub-optimal



1. Strength session likely compromised by residual fatigue from type 5 HIIT (SIT) session performed on the PM the day prior.
2. This residual fatigue is also likely worsened by prior high-volume LIT session.
3. Strength session likely compromised by residual fatigue from prior type 4 HIIT session.



1. Type 5 HIIT session swapped with a type 3 HIIT session to reduce the likelihood of residual fatigue prior to strength session the following day.
2. Strength session moved to >6 h after previous conditioning session to further limit residual fatigue.
3. Strength session moved to AM to avoid residual fatigue induced by type 4 HIIT session in PM.

Note: It should be considered that the movement of strength sessions as per points 2 and 4 may influence subsequent HIIT performance on each day, and depending on various factors (e.g., athlete training status, specific nature of the strength sessions), this influence could be positive or negative.

Figure 6.5 Endurance sport concurrent training program puzzle example.

Additional Nontraining-Related Strategies to Mitigate the Interference Effect

As discussed initially in this chapter, the interference effect is likely a result of compromised performance and/or adaptive responses to individual strength training sessions, culminating in attenuated strength training adaptations. Aside from the manipulation of training variables and weekly session organization, there is potential for other nontraining-related strategies to somewhat mitigate any potential negative influence of HIIT on strength training performance and/or post-exercise adaptive responses during concurrent training.

Strategies to Minimize (Acute) Residual Fatigue Prior to Strength Training

The links between HIIT-induced neuromuscular fatigue and the resulting potential for subsequent reductions in strength training performance suggest interventions aimed at attenuating post-exercise neuromuscular fatigue may be beneficial to limit any potential compromise to strength training performance during concurrent training. There is indeed evidence that certain post-exercise recovery modalities, including cold water immersion (CWI) (94) and dynamic pneumatic compression devices (90), can enhance recovery of the declines in muscle strength or increases in muscle soreness after strenuous exercise. However, whether the implementation of recovery modalities following metabolic conditioning can minimize detrimental effects on subsequent strength training performance has not been investigated. While strategies such as cryotherapy have shown promise for accelerating post-exercise recovery (94), one potential downside of cold therapies applied in the hours prior to strength training sessions are potential short-term reductions in force production or rate of force development due to cold-induced decreases in nerve conduction velocity. As such, there is potential for the application of cryotherapy modalities in close proximity to a subsequent strength training session to actually compromise, rather than enhance, performance. It is therefore possible a certain recovery window after application of post-exercise recovery techniques is necessary to allow for an attenuation of neuromuscular fatigue, coupled with a restoration of neuro-

muscular properties (such as nerve conduction velocities), before subsequent strength training is performed. Another potential concern is that the application of CWI may itself interfere with strength training adaptations, as performing CWI immediately (i.e., within 5 min) after strength training sessions has been shown to blunt maximal strength and muscle size improvements after 12 wk of training (85). Whether this negative effect of CWI on strength training adaptations is apparent only when applied immediately following strength training sessions, as opposed to following metabolic conditioning sessions performed at least 6 h prior to strength training, is currently unclear. More work is needed to better understand how the implementation of post-exercise recovery strategies can influence exercise performance and adaptation to concurrent training, and this is indeed a promising area for future research.

Nutritional strategies also have the potential to influence exercise performance, and therefore may be useful for countering residual fatigue during periods of concurrent training. Apart from maximizing post-exercise adaptive responses (particularly anabolism following strength training), nutrient availability between training sessions (e.g., between a morning metabolic conditioning session and an afternoon strength training session) also likely impacts exercise performance during the second session and, likely, the resulting long-term adaptations. Given the links between muscle glycogen content and endurance performance (12), post-exercise glycogen replenishment between concurrent training sessions performed within the same day (or even on alternate days) may play a role in maximizing exercise performance during the second session. It is important to note that glycogen repletion may be further accelerated when carbohydrate is coingested with protein (16), providing another rationale for consuming protein after completion of metabolic conditioning sessions. As such, the ingestion of protein after both metabolic conditioning and strength training sessions is recommended, with carbohydrate (CHO) potentially provided between sessions to assist with muscle glycogen restoration that may influence performance during subsequent training sessions (irrespective of the focus of the second session, i.e., strength training or HIIT).

As well as macronutrient intake, other ergogenic aids, including caffeine (25) and creatine monohydrate (58), can enhance high-intensity exercise performance and may be useful to counter any residual

fatigue induced by prior metabolic conditioning during concurrent training (see also related section on caffeine in chapter 4). In one study (27), creatine supplementation prevented the impairment to lower-body strength training performance seen following a bout of HIIT (5 km run performed intermittently at 1 min at $\dot{V}O_{2\max}$ /1 min passive recovery). While these findings are promising, further studies are needed to confirm whether the short-term benefits of creatine supplementation translate into improved adaptation (and less interference to strength training adaptations) following longer-term concurrent training. It also remains to be determined whether caffeine supplementation might be effective for countering residual fatigue induced by a morning (e.g., metabolic conditioning) session and subsequently improving performance in an afternoon (e.g., strength training) session (see chapter 4).

Strategies to Enhance Adaptive (Chronic) Responses to Concurrent Training

Since training adaptations are largely underpinned by the accumulation of adaptive responses to single exercise bouts performed during a training program (45, 81), strategies to maximize these responses are paramount for maximizing training adaptation. Although direct evidence is currently lacking, the notion that metabolic conditioning may interfere with adaptive responses to strength training (45) also highlights the relevance of optimizing post-exercise adaptive responses during concurrent training.

Nutrient availability is a key nontraining-related factor for optimizing adaptation to both endurance and resistance training, and concurrent training is naturally no different (73, 80). Nutritional status can influence both exercise performance (as with CHO availability or caffeine ingestion, for example, which can influence exercise capacity and therefore the quality of the training stimulus itself; see chapter 4), as well as the adaptive responses to exercise (such as when training with low CHO availability or with post-exercise protein intake, both of which influence adaptive responses to exercise in muscle).

Nutrient and energy availability clearly have a profound effect on adaptive responses to both strength training and metabolic conditioning, and in some respects, in a divergent manner. In general, adaptive responses to metabolic conditioning seem to be amplified in an energy-restricted environment, whereas an energy-rich environment may be required

to support anabolism following strength training (42). For example, post-exercise adaptive responses related to mitochondrial biogenesis and substrate metabolism are further enhanced when metabolic conditioning is performed under conditions of low energy (in particular, carbohydrate) availability (8). Interestingly, however, anabolic responses to strength training are no different when performed in a glycogen-depleted state (19), whereas the role of protein (in particular, essential amino acid) ingestion in maximizing post-exercise rates of muscle protein synthesis (97) and subsequently lean mass accretion (75) is well established. It should also be mentioned that post-exercise carbohydrate ingestion does not influence muscle anabolism above protein intake alone (40, 93), suggesting there may be limited need to ingest large amounts of CHO following strength training sessions (at least when not followed by HIIT).

Given that nutrient availability plays a key role in adaptation to both metabolic conditioning and strength training, it likely also influences the interference effect with concurrent training. The fact that HIIT can impart significant energy stress on the active musculature, including glycogen depletion and increased AMPK activity (3), potentially provides a suboptimal environment for anabolic responses to strength training (45). Although there is a lack of evidence for glycogen depletion (19) or increased AMPK activity (3) inhibiting anabolic responses to strength training, post-exercise protein ingestion is nonetheless important for promoting protein turnover and remodeling following both metabolic conditioning (73) and strength training (75). Prolonged metabolic conditioning also promotes increased oxidation rates of amino acids (72) derived from the catabolism of muscle proteins, which can result in a negative muscle and whole body protein balance during exercise (52). Protein ingestion following metabolic conditioning sessions is therefore likely to assist in restoring any disruption caused to net protein balance that may limit chronic lean mass accretion during concurrent training.

Clearly, optimizing nutritional strategies (and protein intake in particular) is an important consideration when undertaking concurrent training. Both the amount (i.e., ~1.2–1.7 g/kg body mass per day, or 20 g in a single dose) and pattern (i.e., 4–5 separate intakes of an optimal protein dose over a 12 h period (4)) of protein intake seem important for maximizing the beneficial effects on muscle protein synthesis and/or muscle growth. While 20 g of

protein is often considered an optimal dose, higher protein amounts (e.g., 40 g) might be necessary to maximally stimulate muscle protein synthesis following whole-body exercise (67). Higher protein doses (e.g., ~27.5-40 g) may also be necessary to ingest prior to sleep to maximize overnight levels of muscle protein synthesis (84) and subsequently muscle growth (91). Adoption of these protein intake strategies during periods of concurrent training may potentially minimize any interference effect to strength training adaptations, although empirical evidence for this is currently lacking.

Since the interference effect with concurrent training is likely mediated by either a compromised training quality and/or adaptive response to training, the role of nontraining-related strategies in modulating both of these parameters highlights their potential importance in supporting concurrent training adaptation.

The potential role of these nontraining-related strategies for mitigating the interference effect can be summarized as follows:

- Nutritional strategies (such as creatine and caffeine supplementation), as well as the replenishment of muscle glycogen stores between training sessions, may be useful for attenuating residual (acute) neuromuscular fatigue during concurrent training.
- Other nonnutritional recovery strategies (e.g., cryotherapy) may somewhat attenuate residual (acute) neuromuscular fatigue following metabolic conditioning sessions, which may restore performance during subsequent strength training sessions.
- Protein intake following both metabolic conditioning and strength training sessions is essential for maximizing muscle protein synthesis rates to support necessary muscle remodeling and subsequently (chronic) training adaptation.
- Additional post-exercise carbohydrate intake, while useful for muscle glycogen repletion, does not influence anabolic responses to strength training and may even dampen (chronic) adaptive responses to endurance training in some circumstances.

Summary

Integrating both metabolic conditioning (including HIIT) and strength training into the overall training program presents unique challenges for the practitioner. That the residual effects of each training modality likely influence performance in any subsequent modalities highlights the importance of weekly session sequencing when integrating multiple training modes into the program. Although our understanding of how prescription-related factors may exacerbate any interference effect is improving, more work is required to further characterize these factors, so that practical strategies to mitigate any negative interactions between metabolic conditioning and strength training may be informed.

In the context of the HIIT programming puzzle (figure 1.5), the practitioner requires an understanding of factors that modulate the demands of HIIT sessions, particularly neuromuscular strain, as HIIT formats characterized by large neuromuscular demands are more likely to compromise subsequent strength-training performance. The practitioner should also be mindful of the influence of strength training-induced fatigue influencing the quality of key HIIT sessions, of which sessions with larger neuromuscular demands may likewise suffer most. In either case, the practitioner may need to prioritize performance in key HIIT sessions over strength-training sessions (and vice versa) during different stages of the overall program. Although empirical evidence is lacking, such a periodized approach may help to minimize negative interactions between sessions within the weekly program, which may transfer to better long-term adaptations. Aside from the manipulation of training-related variables, the potential benefits of nutritional interventions and post-exercise recovery strategies for maintaining between-session performance may also aid in mitigating the interference effect; however, evidence for their efficacy in concurrent training scenarios is limited. Together, the integration of metabolic conditioning with strength training clearly forms another complex piece of the programming puzzle, which practitioners must solve within their limitations in an attempt maximize the simultaneous development of strength and endurance capacity in their athletes.

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